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ORIGINAL ARTICLE

IMAGING

Longitudinal Functional Connectome in Pediatric Concussion: An Advancing Concussion Assessment in Pediatrics Study

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Abstract

Advanced magnetic resonance imaging (MRI) techniques indicate that concussion (i.e., mild traumatic brain injury) disrupts brain structure and function in children. However, the functional connectivity of brain regions within global and local networks (i.e., functional connectome) is poorly understood in pediatric concussion. This prospective, longitudinal study addressed this gap using data from the largest neuroimaging study of pediatric concussion to date to study the functional connectome longitudinally after concussion as compared with mild orthopedic injury (OI). Children and adolescents ($n = 967$) 8–16.99 years with concussion or mild OI were recruited from pediatric emergency departments within 48 h post-injury. Pre-injury and 1-month post-injury symptom ratings were used to classify concussion with or without persistent symptoms based on reliable change. Subjects completed a post-acute (2–33 days) and chronic (3 or 6 months via random assignment) MRI scan. Graph theory metrics were derived from 918 resting-state functional MRI scans in 585 children (386 concussion/199 OI). Linear mixed-effects modeling was performed to assess group differences over time, correcting for multiple comparisons. Relative to OI, the global clustering coefficient was reduced at 3 months post-injury in older children with concussion and in females with concussion and persistent symptoms. Time post-injury and sex moderated group differences in local (regional) network metrics of several brain regions, including degree centrality, efficiency, and clustering coefficient of the angular gyrus, calcarine fissure, cuneus, and inferior occipital, lingual, middle occipital, post-central, and superior occipital gyrus. Relative to OI, degree centrality and nodal efficiency were reduced post- acutely, and nodal efficiency and clustering coefficient were reduced chronically after concussion (i.e., at

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3 and 6 months post-injury in females; at 6 months post-injury in males). Functional network alterations were more robust and widespread chronically as opposed to post-acute after concussion, and varied by sex, age, and symptom recovery at 1-month post-injury. Local network segregation reductions emerged globally (across the whole brain network) in older children and in females with poor recovery chronically after concussion. Reduced functioning between neighboring regions could negatively disrupt specialized information processing. Local network metric alterations were demonstrated in several posterior regions that are involved in vision and attention after concussion relative to OI. This indicates that functioning of superior parietal and occipital regions could be particularly susceptible to the effects of concussion. Moreover, those regional alterations were especially apparent at later time periods post-injury, emerging after post-concussive symptoms resolved in most and persisted up to 6 months post-injury, and differed by biological sex. This indicates that neurobiological changes continue to occur up to 6 months after pediatric concussion, although changes emerge earlier in females than in males. Changes could reflect neural compensation mechanisms.

Keywords: children; concussion; functional connectivity; graph theory; orthopedic injury; resting state functional MRI

Introduction

The median annual incidence of traumatic brain injury (TBI) in children is estimated to be 691 per 100,000 population treated in emergency departments. Most injuries are classified as mild in severity (i.e., concussion).^{1,2} Concussion can cause clinical symptoms, including physical, cognitive, and emotional problems, that typically resolve by 1-month post-injury.^{3,4} However, 15–25% of affected children experience persistent symptoms up to ≥ 6 months. Clinical diagnosis of concussion and identification of children at high risk of persistent symptoms post-injury is challenged by the (1) lack of sensitive and specific diagnostic and prognostic biomarkers, and (2) limited understanding of the neurobiology of concussion, especially in a developing brain.⁵

Concussion can alter brain structure and function in children.⁵ Advanced magnetic resonance imaging (MRI) techniques have indicated disrupted network communication and information flow between brain regions following concussion in children.^{6,7} Analysis of temporal correlations of spontaneous blood-oxygenation-level-dependent (BOLD) oscillations using resting-state functional magnetic resonance imaging (rs-fMRI) can be used to map functional communication in the brain non-invasively.⁸ In pediatric concussion, rs-fMRI has been used to detect abnormal brain functioning, and also shows potential for predicting clinical outcomes.^{9–11}

Graph theory can be used to map the structural and functional connectivity of brain regions within global and local networks (i.e., connectome).¹² In pediatric concussion, structural connectome alterations can occur, and suggest that white matter alterations can impair subnetwork connectivity and reduce efficiency of information flow.^{6,7,13,14} This has important implications for functional connectivity at both the global and local network levels. However, only one study has examined the func-

tional connectome in a relatively small sample of children with a range of injury severities (i.e., complicated mild, moderate, and severe TBI).¹⁵ The findings of that study included atypical global network organization (i.e., lower modularity, higher small worldness, and lower assortativity), which suggests that pediatric TBI disrupts specialized network processing, the balance between local and global information processing, and connectivity between brain networks. Understanding the functional brain architecture based on graph theory could help to increase knowledge about neural markers that are specific and sensitive to pediatric concussion, because of the ability to examine both global and local (regional) network functioning.¹²

Inconsistencies across studies limit understanding of changes in functional brain networks following concussion in children.¹⁶ Most previous studies used a retrospective or cross-sectional design, included samples with restricted age ranges (e.g., adolescents only) and biased sex ratios (i.e., male > female), and examined specific brain networks (e.g., default mode and salience network).^{10,11,17,18} Comparison groups also vary across studies, with some studies including typical, uninjured children (“healthy controls”) and others including those with extracranial or orthopedic injury (OI). A growing body of literature highlights the importance of comparison group selection in neuroimaging research on concussion, indicating that children with concussion are more similar to those with OI than to healthy controls.^{19,20} Healthy controls can help to determine whether concussion alters typical development, but behavioral²¹ and mounting neuroimaging^{6,22,23} evidence indicates that a healthy control group can inflate group differences for several reasons. For instance, children’s risk of traumatic injury is influenced by parental and sociodemographic factors, as well as by child characteristics related to

impulsivity and risk-taking behaviors.^{24,25} Those pre-injury factors increase children's risk of traumatic injury regardless of whether the injury involves the head or is extracranial.²⁶ Therefore, including an OI group instead of a healthy control group can account for pre-morbid factors related to risk of injury generally. An OI group also can help to account for post-injury factors that can influence clinical outcomes and brain structure and functioning, including the effects of pain, inflammatory response to injury, loss of quality of life with altered engagement in typical routine, and medical interactions.^{27–31} The use of an OI comparison group also accounts for the psychosocial effects of having had a traumatic injury, including the acute emotional effects of trauma (e.g., stress) caused by the event.²⁰ Using a healthy control group means that group differences based on pre-morbid family or individual characteristics, or post-injury factors such as change in routine and systemic inflammation could be misinterpreted as effects specific to a brain injury. Using an OI comparison group helps to control for these effects and therefore presents a more conservative comparison group for determining whether differences are related to the brain injury specifically rather than to pre-morbid or general post-injury factors, providing increased scientific and clinical relevance, than a healthy control group. Indeed, we previously showed no brain differences between children with concussion and those with OI, whereas children with both concussion and OI differed from healthy controls in terms of structural network connectivity.⁶ Another tractography finding in an independent sample of older children and young adults had similar findings.²²

This prospective, longitudinal study used data from the largest MRI study of pediatric concussion and mild OI to date to examine the functional connectome and its trajectory across time post-injury. Specifically, changes between the post-acute (e.g., 2–33 days post-injury) and chronic (i.e., 3 or 6 months post-injury via random assignment) functional connectome of children with concussion was compared with those of children with mild OI. Because the current study aimed to identify differences in the functional connectome that are specific to brain injury and the utility of those differences for prognostication, we believe that the inclusion of a mild OI comparison group provided a more appropriate and conservative comparison than inclusion of a healthy control group. We expected lower integration and greater segregation following pediatric concussion based on previous findings regarding the structural connectome in this sample¹⁴ and an independent sample of pediatric TBI.¹⁹ Prior research supports sex- and age-dependent effects of concussion on clinical outcomes as well as on brain function and structure in children.^{7,13,23,32} Therefore, we investigated the influence of three important moderators on global and local brain metrics: symptom persistence at

1 month post-injury, age at injury, and biological sex. Based on previous findings, including in the current sample, we expected greater chronic alterations after concussion to be demonstrated in females than in males, and also in older children (i.e., adolescents) than in younger (i.e., school-age) children.^{7,13,32}

Methods

Study design and procedure

Data were drawn from the Advancing Concussion Assessment in Pediatrics (A-CAP) study.³³ This multi-site study used a prospective, concurrent cohort design to study outcomes longitudinally in children and adolescents between 8 and 16.99 years of age within 48 h following concussion versus those with mild OI. Study recruitment occurred in the emergency department of five children's hospitals that are all members of the Pediatric Emergency Research Canada (PERC) network^{33,34}: Alberta Children's Hospital (Calgary), Children's Hospital of Eastern Ontario (Ottawa), Centre Hospitalier Universitaire Sainte-Justine (Montreal), Stollery Children's Hospital (Edmonton), and British Columbia Children's Hospital (Vancouver). Enrolled participants at each site returned for three additional follow-up assessments: a post-acute assessment (i.e., targeted for 10 days post-injury; range 2–33 days) and two chronic assessments, at 3 and 6 months post-injury. Overall study attrition rates were 15%, 25%, and 28% for the post-acute, 3-month, and 6-month assessments, respectively, similar to other studies of pediatric concussion. Information about the acute clinical presentation and mechanism of injury, and a demographic questionnaire, were collected at the post-acute follow-up. All eligible participants (i.e., without MRI contraindication; see the following section) completed 3T MRI at the post-acute assessment and were randomly assigned to complete a second MRI scan at 3 or 6 months post-injury.

The study was conducted with the approval of the research ethics board at each study site. All participants provided written informed assent and parents/guardians provided written informed consent.

Participants

Concussion. Children included in the concussion group had sustained a blunt head trauma resulting in at least one of the following three criteria, consistent with the World Health Organization definition of mild TBI³⁵: (1) observed loss of consciousness, (2) Glasgow Coma Scale score of 13–14, or (3) at least one acute sign or symptom of concussion as noted by emergency department medical personnel on a standard case report form, including post-traumatic amnesia, focal neurological deficits, vomiting, headache, dizziness, or other mental status changes. Subjects were excluded if they demonstrated delayed

neurological deterioration (e.g., Glasgow Coma Scale score <13), required neurosurgical intervention, or had loss of consciousness >30 min or post-traumatic amnesia >24 h.³³

Mild OI. Subjects with mild OI sustained an upper or lower extremity fracture, sprain, or strain caused by blunt force trauma, associated with an Abbreviated Injury Scale score ≤4 (i.e., “mild” severity).³⁶ Subjects were excluded from the OI group if they had head trauma, symptoms of concussion, or any injury requiring surgical intervention or procedural sedation.³³

Exclusion criteria. Both injury groups were subject to additional exclusion criteria: any other severe injury as defined by an Abbreviated Injury Scale score >4; hypoxia, hypotension, or shock during or following the injury; previous concussion within 3 months prior or any prior TBI requiring hospitalization; pre-morbid neurological disorder or severe neurodevelopmental disability; injury resulting from non-accidental trauma; severe psychiatric disorder requiring hospitalization within the past year; or contraindications to MRI (e.g., metallic implants, orthodontia).³³

Symptoms

The Health and Behavior Inventory was used to assess cognitive and somatic symptoms. This measure has good internal consistency and test–retest reliability, and has been adopted as a core measure in the Common Data Elements for Pediatric TBI.^{37–39} Total pre-morbid (pre-injury) symptoms were rated by parents during the post-acute visit, and total post-injury symptoms were rated by both parents and children weekly and also at each follow-up assessment.⁴⁰ A reliable change index *z*-score comparing total 1-month post-injury symptom scores to pre-morbid scores was calculated using the following formulas based on regression analyses using data from the OI group for child ratings³⁹:

$$z = \{ \text{post-injury score} \times [3.44 + (0.50 \times \text{pre-morbid score})] \} / 6.89$$

and parent ratings:

$$z = \{ \text{post-injury score} \times [2.32 + (0.52 \times \text{pre-morbid score})] \} / 6.683$$

Results were used to classify children with concussion into two groups using a critical *z*-score >1.65 (one-tailed *p*<0.05): (1) with persistent symptoms (significant increase at 1-month post-injury relative to pre-morbid) and (2) without persistent symptoms (no significant increase at 1-month post-injury relative to pre-morbid).^{3,4}

MRI

Eligible participants completed at least one 3T MRI scan without sedation, which included T1-weighted and

rs-fMRI sequences. T1-weighted images were acquired with 0.8 mm isotropic resolution at all sites using fast spoiled gradient echo (FSPGR) BRAVO sequences [General Electric: repetition time (TR)/echo time (TE)/inversion time (TI)=7.9, 8.25/3.6/600 ms, 4:47 min; Siemens: Magnetization-Prepared Rapid Acquisition Gradient Echo (MPRAGE) with TR/TE/TI=1880, 2200/2.5, 2.9/948 ms, 5:28 min], and rs-fMRI data was acquired using a single shot gradient echo planar imaging (EPI) sequence with a scan duration of 8 min and 10 sec [TR=2000 ms, TE=30 ms, field of view (FOV)=23 cm, 36 slices, voxel dimensions=3.6 mm³]. At all sites, children were asked to keep their eyes open, not think of anything, and look at a cross presented on a screen. Foam padding was used to minimize head motion and headphones were used to minimize noise.³³

Quality assurance. T1-weighted and rs-fMRI DICOM[®] data were converted into NIfTI format using the dcm2niix tool in MRICron (<https://github.com/rordenlab/dcm2niix>). Initial visual quality assurance of raw images was conducted for both image types to identify and exclude scans that demonstrated structural abnormalities at the time of the scan, scanner artifact such as aliasing or warping, incomplete acquisition, or reduced FOV (i.e., cropped), or that were not collected using the default scan parameters.

Image processing. Images were pre-processed and processed using a Nipype implementation of commands from the Advanced Normalization Tools (ANTs) v2.2.0,⁴¹ FMRIB Software Library (FSL) v5.0.9,⁴² and Analysis of Functional Neuroimages (AFNI) v.20160207.⁴³

The T1-weighted images were corrected for intensity non-uniformity (N4BiasFieldCorrection), skull-stripped using the OASIS30ANTs template (antsBrainExtraction), and normalized using non-linear registration to standard space (i.e., MNIPediatricAsym:cohort-5; antsRegistration).⁴⁴ For the rs-fMRI scans, a deformation field that was estimated based on the fMRIprep fieldmap-less approach was used to correct for susceptibility distortions in the rs-fMRI data.⁴⁵ This approach applies a deformation field that was calculated during co-registration (implemented using ANTs antsRegistration.sh using constrained non-zero deformation along the phase encoding direction with modulation of the average field map template)⁴⁶ of each functional image to the reference T1-weighted image for each scan.⁴⁷ Each functional image was co-registered to the T1-weighted image using FSL FLIRT⁴⁸ with the boundary-based registration cost-function.⁴⁹ Nine degrees of freedom were used to configure the co-registration to account for any remaining distortions in the functional image. Functional images were then slice-time corrected using AFNI 3dTshift.⁵⁰

Transformation matrices (i.e., six corresponding rotation and translation parameters associated with in-scanner head motion) were estimated for each functional image before spatiotemporal filtering using FSL MCFLIRT.⁵¹ Framewise displacement (FD) was calculated in FSL motion outliers.^{51,52} Functional images were slice-time corrected using AFNI 3dTshift.⁵⁰ To minimize the smoothing effects of other kernels, each function image was warped in template space with a single interpolation step by composing all of the transformation parameters (i.e., transform matrices, susceptibility distortion correction, co-registrations to T1-weighted template space), using the Lanczos interpolation configuration in ANTs antsApplyTransforms.⁵³

Denoising. Each functional run was denoised using AFNI's 3dTproject, accounting for the following regressors⁵⁴: (1) six motion parameters (3 rotations and translations in x y and z directions); (2) global, white matter and cerebrospinal fluid signal; (3) derivatives, squares and squared derivatives of each noise regressor; (4) FD motion spikes (see subsequent description), and (5) linear and polynomial trends. All regressors were detrended by regressing out linear and polynomial trends. A whole-brain mask was generated and used to extract global signals. Next, to ensure that signals pertained to gray matter, white matter and cerebrospinal fluid signals were extracted from eroded segmentation masks.⁵⁵

Head motion was estimated using the FD relative root mean squared (RMS),⁵⁶ which was calculated during time-series realignment using FSL MCFLIRT.⁵² Motion spikes were defined as FD threshold ≥ 0.25 mm⁵² and were included in the model as dummy variables (i.e., spike regression).⁵⁷ High-pass filtering at a threshold > 0.01 Hz was applied in the same model to account for low frequencies; no high-frequency threshold was applied for temporal filtering to preserve connectivity information at higher frequencies.^{58–62} This can improve the stability of pairwise connections and maintain similar effects to bandpass filtering for motion mitigation.⁶³ Scans were excluded for gross motion if they had an average FD RMS ≥ 0.20 mm or ≥ 60 (25%) volumes with FD RMS ≥ 0.25 mm.⁵⁷

Network construction

The 90 supratentorial region Automated Anatomical Label atlas (AAL-90) was used to identify 90 nodes (brain regions).⁶⁴ Initially, the AAL-90 atlas was co-registered to the pediatric template (antsregistration.sh, with multilabel interpolation). The labeled template was then used to identify and extract the corresponding time series for each node ($n=90$) from the de-noised functional images. Significant signal dropout ($> 50\%$ of the voxels) was demonstrated in the bilateral gyrus rectus

for 154 (16.8%) scans and in other (non-overlapping) nodes for 15 scans. Therefore, the bilateral gyrus rectus was excluded as a region of interest, and the other 15 scans were excluded from subsequent analysis. The resulting time series (with 88 nodes) were then used to construct 88×88 weighted connectivity matrices using Pearson correlation.

To factor out the variability in network density across scans^{65,66} and improve the stability of graph theoretical measures,⁶⁷ a proportional threshold was applied to the weighted matrices. Percolation analysis⁶⁸ was used to identify the threshold that removed the weakest connections while ensuring that matrices were fully connected.^{69,70} For this approach, multiple proportional thresholds (i.e., values between 0.01 and 1 at a step of 0.01) are applied serially to identify the value at which the matrices become fragmented. Percolation analysis was used because of the lack of consensus in the literature for threshold definition, and to avoid network fragmentation across scans.⁶⁹ A proportional threshold of 0.25 preserved fully connected matrices in 98% of the scans. Fragmented scans were excluded from the current analysis ($n=20$).

Network metrics

Graph theoretical metrics based on the absolute weighted connectivity matrix of each scan and 1000 randomly generated networks were calculated in MATLAB using the GRaph thEoreticAl Network Analysis (GRETNA) toolbox v.2.0.⁷¹ Global and regional (nodal) level metrics included clustering coefficient, small-worldness, efficiency, betweenness centrality, and degree centrality.^{71,72} Each metric definition and description is provided in Table 1.

Multi-scanner harmonization

Prior to final analysis, network metrics were harmonized for site (scanner) differences using ComBAT in RStudio v1.1.383 (R v4.0.3).^{73–75} Group, time (days) post-injury, age at injury, and sex were included in the covariate matrix during harmonization, and model fitting was conducted with participant as a random effect. We previously validated this approach for network analysis.⁷⁶

Statistical analysis

Demographic data were analyzed using *t* tests for continuous variables and χ^2 techniques for categorical variables.

Multiple linear mixed-effects models were computed in RStudio using the lmerTest package^{7,74,75,77,78} to investigate the relations of group (concussion, OI), the linear and quadratic effects of time (days) post-injury, age at injury, sex, and group by time by age, group by time by sex, and their lower-order interactions on each harmonized network metric, controlling for the random

Table 1. Graph Theory Metrics for Global and Local Network Levels

<i>Metric</i>	<i>Abbreviation (global/nodal)</i>	<i>Definition</i>	<i>Interpretation</i>	<i>Classification</i>
Efficiency	Eg/Ne	Strength of parallel information transfer among each node and all other nodes within the whole graph	Measure of integration, with higher efficiency indicating greater integration (via shorter path lengths) among a select brain region and other regions within the network	Integration
Clustering coefficient	Cp/NCp	Average proportion of node neighbors that are also considered to be neighbors	Overall extent to which proximal network regions are connected or clustered	Segregation
Small-worldness	Sigma (σ)	Ratio between the proportional clustering coefficient and shortest path length relative to those of randomly generated networks	Index of brain connectivity, or the presence of highly specialized and segregated regions, necessary for functional specialization, that are strongly connected to facilitate efficient integration and global network processing	
Betweenness centrality	Bc	Frequency that a node is part of average shortest paths		Centrality
Degree centrality	Dc	Number of edges (connections) of each node		Centrality

Definitions of global and local network parameters based on Rubinov and Sporns, 2010.⁷²

effect of participant. Hemisphere did not moderate group differences in preliminary analyses. Therefore, only the main effect of hemisphere was included in each model. This approach was repeated to compare functional network metrics among symptom status groups (i.e., concussion with persistent symptoms, concussion without persistent symptoms) and OI. The final model is given by the following formulas for global and nodal metrics, respectively:

Harmonized global metric $\sim \text{Group} \times (\text{Time} + \text{Time}^2) \times (\text{Age} + \text{Sex}) + (1 | \text{Participant})$
or

Harmonized regional (nodal) metric $\sim \text{Group} \times (\text{Time} + \text{Time}^2) \times (\text{Age} + \text{Sex}) + \text{Hemisphere} + (1 | \text{Participant})$

To correct for multiple comparisons, the false discovery rate (FDR) at corrected threshold of $p < 0.05$ for injury groups and $p < 0.025$ for symptom groups (based on parent and children report) was used for local (regional) metrics to account for 44 regions bilaterally.⁷⁹ For follow-up analyses of all significant (i.e., survived correction for multiple comparisons) group effects or group-interaction terms, standardized effect size (i.e., Cohen's d) was assessed for group differences within the context of the final model for each network metric, with small, medium, and large effect size indicated by $|0.20| \leq d < |0.50|$, $|0.50| \leq d < |0.80|$, and $d \geq |0.80|$.⁸⁰ For follow-up of significant group differences, group means were modeled within the respective context of the average days post-injury at each assessment for interactions with time post-injury (i.e., 10, 90, 180 days post-injury), in male and female children for interactions with sex, and in younger (i.e., 10th percentile age at injury) and older (90th percentile age at injury) for interactions with age at injury (see Fig. S1). Only effects with a 95% confidence interval range for d that excluded 0 were considered robust, and are described subsequently.

The BrainNet Viewer toolbox in MATLAB R2019a was used to display nodal level results.⁸¹

Data availability

A dataset with deidentified participant data and a data dictionary will be made available upon reasonable request from any qualified investigator, subject to a signed data access agreement, with publication.

Results

Sample

Information about the overall A-CAP study sample and the derivation of the current sample is summarized in Figure 1. Overall, of 3075 eligible children, 967 consented to participate, and 846 returned for at least one assessment. The A-CAP study attrition rates were similar to that in other pediatric concussion studies (post-acute: 15%; 3 months: 25%; 6 months: 28%).⁸² Children who returned at follow-ups did not differ significantly from those who did not return in age, sex, race, or parental education, except at 6 months, when children who returned had higher parental education than those who did not return. A total of 671 children completed at least one MRI, for a total of 1144 completed scans. Orthodontia and scheduling difficulties were the most common reasons that MRI was not completed. Children who completed an MRI were slightly younger (mean [M]=12.23, standard deviation [SD]=2.38 years; $t=5.29$, $p < 0.001$) and were more often male (402 male/266 female; $\chi^2=6.22$, $p=0.013$), but were similar in race and parental education relative to children who did not complete an MRI (M=13.29, SD=2.37 years; 88 male/90 female). Of the children who completed MRI scanning, 661 had at least one usable rs-fMRI scan, resulting in a total of 1130 rs-fMRI scans before quality assurance. Of those, 192 (17%; 131 concussion/61 OI) scans were excluded after quality checks as follows: 104 (9%; 72 concussion/32 OI) for gross motion; 35 (3%; 23 concussion/12 OI) for poor registration, 34 (3%; 25 concussion/9 OI) for incomplete or wrong acquisition; 15 for signal dropout (1.3%, 10 concussion/5 OI); and 4 (0.3%; 1

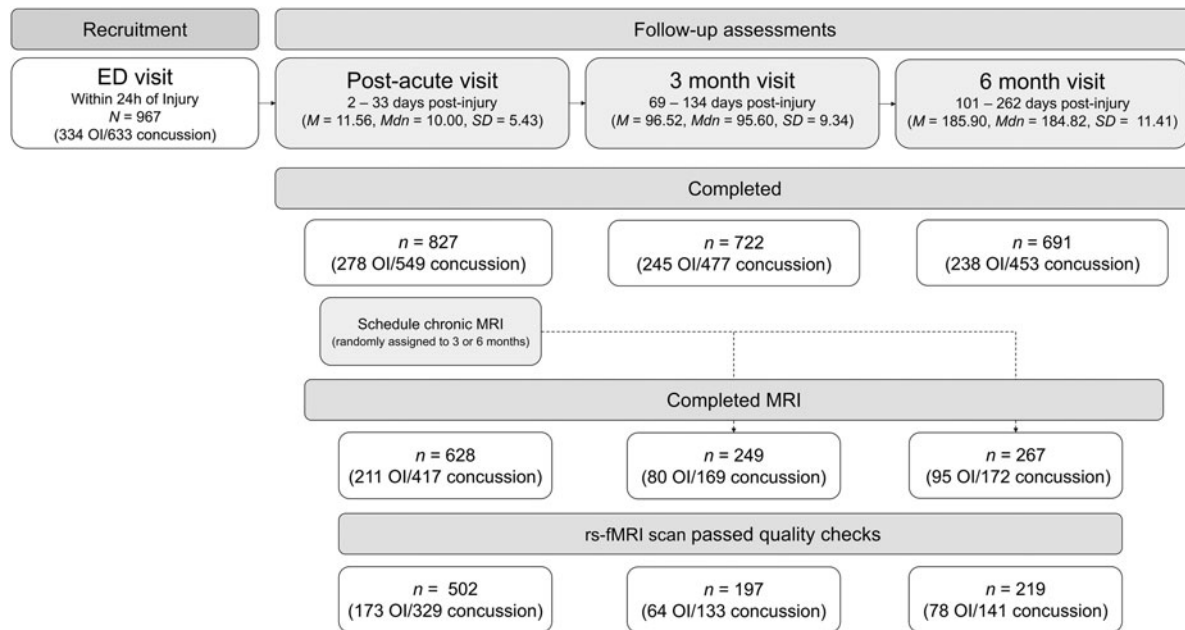


FIG. 1. Summary data for the overall Advancing Concussion Assessment in Pediatrics (A-CAP) study sample and the derivation of the current sample. Overall, of 3075 eligible children, 967 consented to participate, and 846 returned for at least one assessment. The A-CAP study attrition rates were similar to in other pediatric concussion studies (post-acute: 15%; 3 months: 25%; 6 months: 28%). A total of 671 children completed at least one magnetic resonance imaging (MRI) scan, for a total of 1144 completed MRI scans. Orthodontia and scheduling difficulties were the most common reasons that MRI was not completed. Of the children who completed MRI, 661 completed at least one resting state functional magnetic resonance imaging (rs-fMRI) scan, for a total of 1130 scans. Of those, we excluded 192 (17%; 131 concussion/61 orthopedic injury [OI]) scans after quality checks, and 20 scans that had fragmented network topology (i.e., not fully connected). Therefore, the final sample and analyses included 918 scans from 585 children (386 concussion/199 OI).

concussion/3 OI) for structural abnormalities. Additionally, 20 scans (2%; 12 concussion/8 OI) were excluded because the network topology was fragmented following thresholding. Therefore, the final data set included 918 scans (containing both T1-weighted and rs-fMRI) from 585 children (post-acute $n=502$; 3 months post-injury $n=197$; 6 months post-injury $n=219$).

Demographic and injury characteristics

The final sample of children with concussion did not differ from those with mild OI in age at time of injury, sex, parental education, race, average (i.e., in-scanner motion), or whether the injury occurred during sport/recreation, but did differ in terms of injury mechanism (see Table 2). For children with concussion, those with persistent symptoms 1 month post-injury (based on child and parent report) were older, more commonly female, and differed in parental education from those without persistent symptoms (Table S1).

Global network metrics

Injury groups. Group differences in global clustering coefficient (C_p) were moderated by age at injury and time post-injury, which was lower 3 months after concussion relative to OI in older children (Fig. 2A; for statistical results see Fig. S1 and Table S2).

Persistent symptom groups. Time post-injury and sex moderated differences in global clustering coefficients among persistent symptom groups based on parent report (Fig. 2B; for statistical results see Table S3). Clustering coefficient was lower 3 months after concussion in girls with persistent symptoms post-injury than in girls with OI.

Global metrics did not differ among persistent symptom groups based on child report.

Regional and local network metrics

Results for models examining group differences in regional and local network metrics are summarized in Figure 3, and statistical results are reported in Figure S1.

Table 2. Sample Demographic and Injury Characteristics

		Concussion n = 386	Orthopedic injury n = 199	p-value
Study site/scanner (n [%])				0.002
	Calgary-GE	90 (23.3)	40 (20.1)	
	Edmonton-Prisma	80 (20.7)	41 (20.6)	
	Montreal-GE	23 (6.0)	5 (2.5)	
	Montreal-Prisma	14 (3.6)	7 (3.5)	
	Ottawa-Skyra	95 (24.6)	33 (16.6)	
	Vancouver-GE	84 (21.8)	73 (36.7)	
Age (Mean [SD] years)		12.34 (2.41)	12.50 (2.2)	0.448
Sex (n [%] male)		240 (62.2)	111 (55.8)	0.159
Parental education (n [%])				0.953
	No certificate, diploma or degree	10 (2.6)	3 (1.5)	
	High school diploma or equivalent	48 (12.4)	25 (12.6)	
	Trades certificate or diploma	32 (8.3)	16 (8.0)	
	2-year college diploma	70 (18.1)	39 (19.6)	
	4-year bachelor's degree	139 (36.0)	63 (31.7)	
	Master's degree	43 (11.1)	26 (13.1)	
	Doctoral degree (PhD or similar)	12 (3.1)	6 (3.0)	
	Medical degree	6 (1.6)	4 (2.0)	
Race (n [%] white)		268 (69.4)	138 (69.3)	0.427
Mechanism of injury (n [%])				<0.001
	Fall	127 (32.9)	80 (40.2)	
	Struck object	85 (22.0)	33 (16.6)	
	Struck person	58 (15.0)	20 (10.1)	
	Bicycle related	6 (1.6)	9 (4.5)	
	Motor vehicle collision	3 (1.0)	0 (0.0)	
	Other	4 (1.0)	12 (6.0)	
	Unknown	6 (1.6)	11 (5.5)	
Sport-related injury (n [%] sport/recreational play)		248 (64.2)	136 (68.3)	0.493
Frame-wise displacement (FD; mean [SD])		0.08 (0.04)	0.08 (0.04)	0.064

GE, General Electric; SD, standard deviation.

Injury groups. Differences in regional local network metrics between the injury groups are illustrated in Figure 4 (for statistical results see Table S2).

Time post-injury moderated group differences for superior occipital gyrus degree centrality (Dc), superior parietal gyrus efficiency (Ne), and clustering coefficient (NCp) of the fusiform gyrus (Fig. 4A). Follow-up tests indicated that superior occipital gyrus Dc was lower post-acutely after concussion relative to OI; and superior parietal gyrus Ne and fusiform gyrus NCp was lower 3 months after concussion relative to OI.

Time post-injury and sex moderated group differences in Ne and NCp for several parietal and occipital regions (Fig. 4B). Follow-up tests indicated that Ne was lower in females post-acutely and 3 months after concussion relative to OI for the superior occipital gyrus; and 3 months post-injury for the angular and lingual gyri. Ne of the lingual gyrus also was reduced in males 6 months after concussion relative to OI. In contrast, middle occipital gyrus Ne was higher 6 months after concussion relative to OI in females. NCp of the post-central gyrus was lower 3 months after concussion relative to OI in females.

Persistent symptom groups. No robust differences were observed in local network metrics between symptom groups based on child report. However, time post-

injury and biological sex moderated differences in regional Dc, Ne, and NCp between symptom groups based on parent report (Fig. 5; for statistical results see Table S3).

In females, Dc (Fig. 5A) was lower 3 months post-injury for the middle occipital gyrus and cuneus in concussion with persistent symptoms relative to concussion without persistent symptoms, and for the lingual gyrus in concussion with persistent symptoms relative to OI. At 6 months post-injury, Dc was greater in concussion with persistent symptoms than in OI for the lingual gyrus, and in the middle occipital gyrus in concussion without persistent symptoms in females. In males, Dc of the cuneus was lowest in concussion with persistent symptoms relative to concussion without persistent symptoms and OI 6 months post-injury.

In females, Ne (Fig. 5B) was lower for the superior occipital gyrus, calcarine fissure, cuneus, and lingual gyrus in concussion without persistent symptoms than in OI post-acutely. At 3 months post-injury, Ne was lower for the superior occipital gyrus, calcarine fissure, cuneus, lingual gyrus, inferior occipital gyrus, angular gyrus, and middle occipital gyrus in concussion with persistent symptoms than in concussion without persistent symptoms or OI. The opposite pattern was observed in males, whereby Ne was higher for the inferior occipital gyrus in concussion with persistent symptoms than in concussion without persistent symptoms and OI 3 months

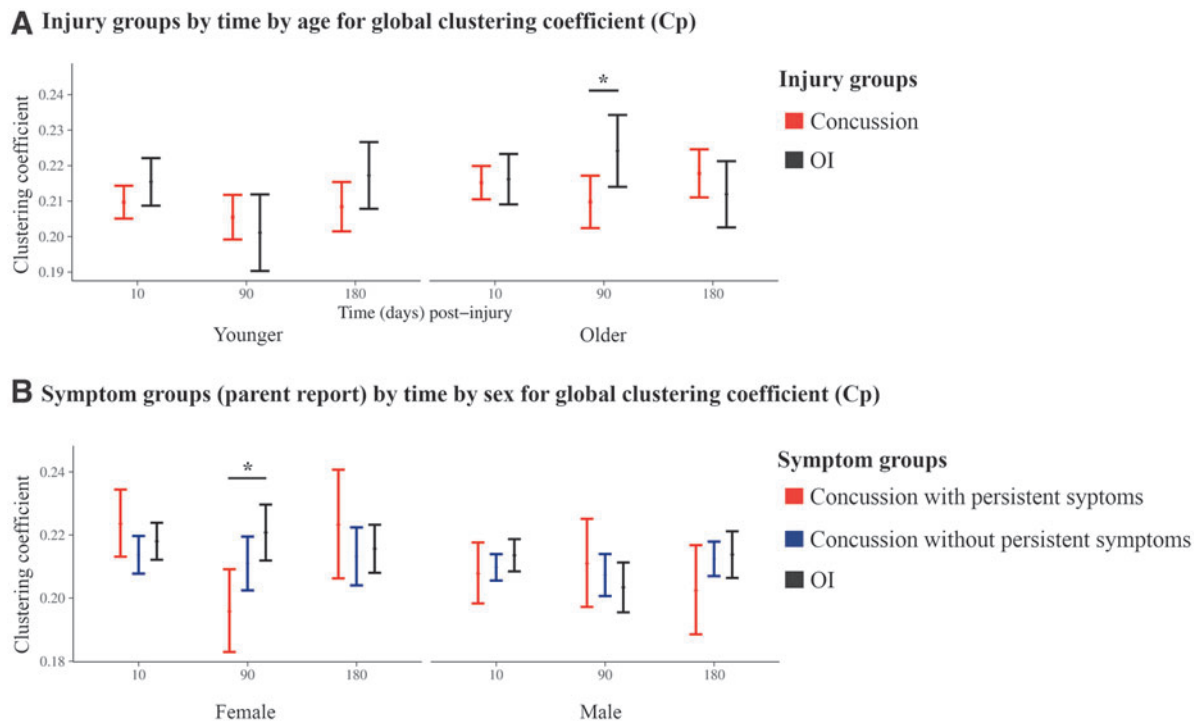


FIG. 2. Group differences in global network metrics were moderated by time post-injury, age at injury, and sex. Graphs illustrate the moderating effects of **(A)** time post-injury and age for global clustering coefficient (Cp) between the children with concussion and those with mild orthopedic injury (OI), and **(B)** time post-injury and biological sex for differences in global clustering coefficient (Cp) (top) and efficiency (Eg) among those with concussion with or without persistent symptoms (based on parent-reported symptoms at 1 month post-injury) and OI. *Robust group differences (i.e., Cohen's d 95% confidence interval excluded 0).

post-injury. At 6 months post-injury, Ne was higher for the calcarine fissure in concussion with persistent symptoms than in OI, and for the middle occipital gyrus in concussion without persistent symptoms than in OI in females. The opposite pattern was seen in males, with lower calcarine fissure and cuneus Ne demonstrated 6 months post-injury in concussion with persistent symptoms than in OI.

NCp (Fig. 5C) was lower 3 months post-injury in concussion with persistent symptoms for the calcarine and postcentral gyrus, and in concussion without persistent symptoms for the post-central gyrus, than in OI in females. However, at 6 months post-injury, NCp was higher for the calcarine fissure in concussion with persistent symptoms than in OI in females.

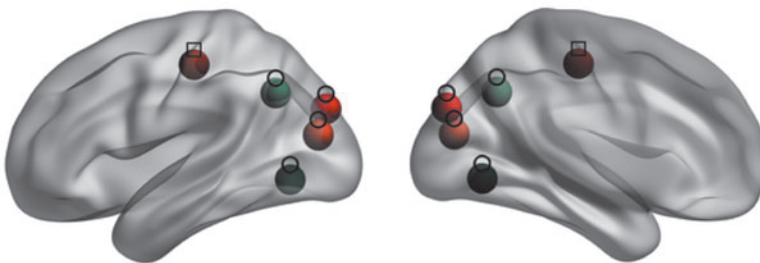
Discussion

In this large, prospective and longitudinal study of the functional connectome in pediatric concussion, abnormalities at both the global and local (regional) brain network level were demonstrated following concussion relative to mild OI. Network differences were moderated

by symptom persistence 1 month post-injury, time post-injury, biological sex, and, to a lesser extent, age at injury. Network metrics distinguished concussion from OI post-acutely (i.e., ~10 days post-injury), although the magnitude and regional specificity of group differences suggest that the results likely have little clinical utility for diagnosis when applied to individual children. Network metrics also demonstrated poor prognostic utility, with no post-acute differences between symptom persistence groups. In contrast to post-acute differences, more robust network alterations emerged 3 months after concussion, especially in females, adolescents, and subjects with slow recovery (i.e., concussion with persistent symptoms at 1 month), and persisted up to 6 months post-injury. Overall, the results suggest that functional brain network changes can occur and persist after most children and adolescents with concussion have recovered, and that concussion is best contextualized while considering time post-injury, symptom persistence, and neurodevelopmental factors, such as biological sex and age at injury. Overall results are

A Injury groups (concussion - OI) by time**Brain regions**

- Angular gyrus
- Calcarine fissure
- Cuneus
- Fusiform gyrus
- Inferior occipital gyrus
- Lingual gyrus
- Middle occipital gyrus
- Postcentral gyrus
- Superior occipital gyrus
- Superior parietal gyrus

B Injury groups (concussion - OI) by time by sex**Metric**

- Efficiency (E_g/N_e)
- Clustering coefficient (C_p/N_Cp)
- △ Degree centrality (D_c)

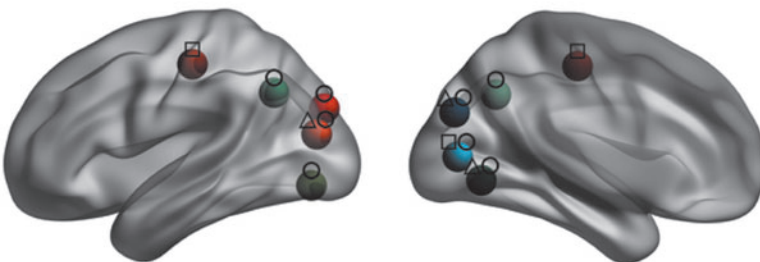
C Symptom groups (parent report) by time by sex

FIG. 3. Summary of results for regional (nodal) graph metrics between the groups for **(A)** injury groups by time, **(B)** injury groups by time by sex, and **(C)** symptom groups based on parent report by time by sex.

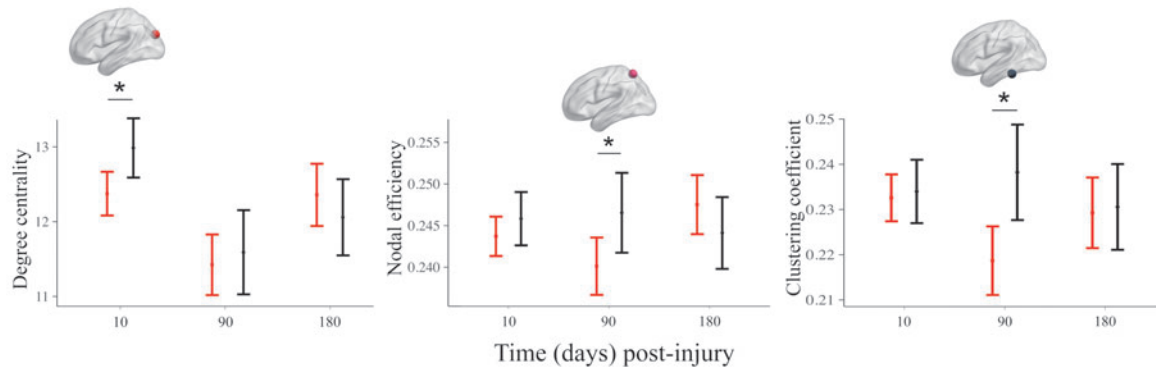
largely consistent with the pattern of structural brain differences demonstrated in this sample.^{7,13,32}

Global clustering coefficient was reduced at 3 months post-injury in older children as well as in females with slow recovery after concussion, as compared with OI, indicating that concussion disrupts global brain functioning through reduced segregation (i.e., less connectivity among regions of local subnetworks that are necessary for specialized information processing). In this sample, we found white matter microstructure trajectory differences in regional and local, but not global, network structure that were similarly moderated by symptom persistence.^{7,13,14} Biologically, those results are consistent with the idea that concussion is associated with subtle, diffuse alterations in white matter microstructure (e.g., diffuse axonal injury). Those alterations may lead to loss of local connections several months after concussion and the disrupted global brain network functioning seen currently, particularly in females

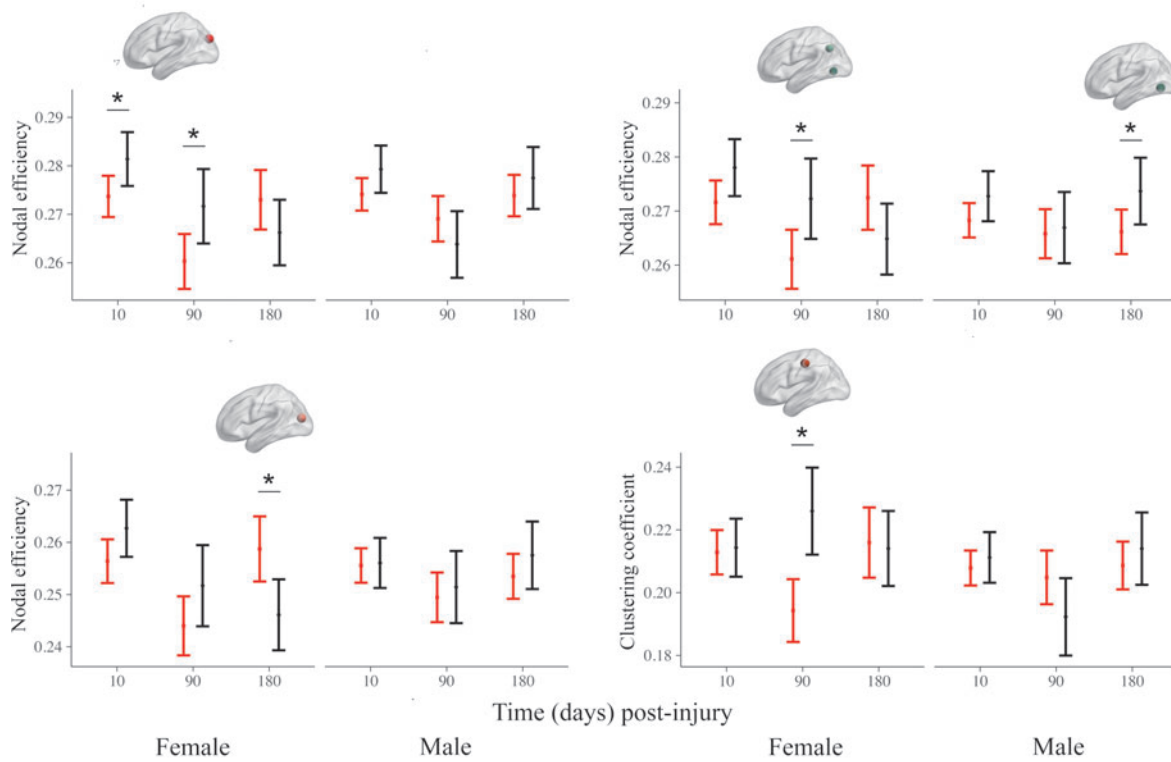
with slow recovery. However, global differences resolved by 6 months post-injury, suggesting that functional changes develop over time following concussion but later resolve. This dependence on time post-injury could help to account for discrepancies with the only other known study of the functional connectome in pediatric TBI, which found altered modularity but not clustering coefficient several years following TBI (across severities).¹⁵

Degree centrality and nodal efficiency were reduced post-acutely after concussion, indicative of regional connection loss and weakening of local networks. However, results were not diagnostic nor prognostic of poor recovery in children with concussion; the small effect sizes, regional specificity, and moderating effects of sex indicate limited clinical application of these functional metrics for predicting clinical outcomes in individual children. Our previous findings in this sample^{6,7,13,14} and independent studies also showed that differences in

A Injury group by time post-injury for degree centrality (Dc), nodal efficiency (Ne), and clustering coefficient (NCp)



B Injury group by time post-injury by sex for nodal efficiency (Ne) and clustering coefficient (NCp)



Injury groups

- Concussion
- OI

Brain regions

- Angular gyrus
- Fusiform gyrus
- Lingual gyrus
- Middle occipital gyrus
- Postcentral gyrus
- Superior occipital gyrus
- Superior parietal gyrus

FIG. 4. Differences in regional (nodal) graph metrics between the injury groups were moderated by time post-injury and sex. Graphs illustrate the moderating effects of **(A)** time post-injury for degree centrality (Dc), nodal efficiency (Ne), and clustering coefficient (NCp), and **(B)** time post-injury by sex for Ne and clustering coefficient (NCp). *Cohen's d indicative of robust effects (i.e., d 95% confidence interval excluded 0).

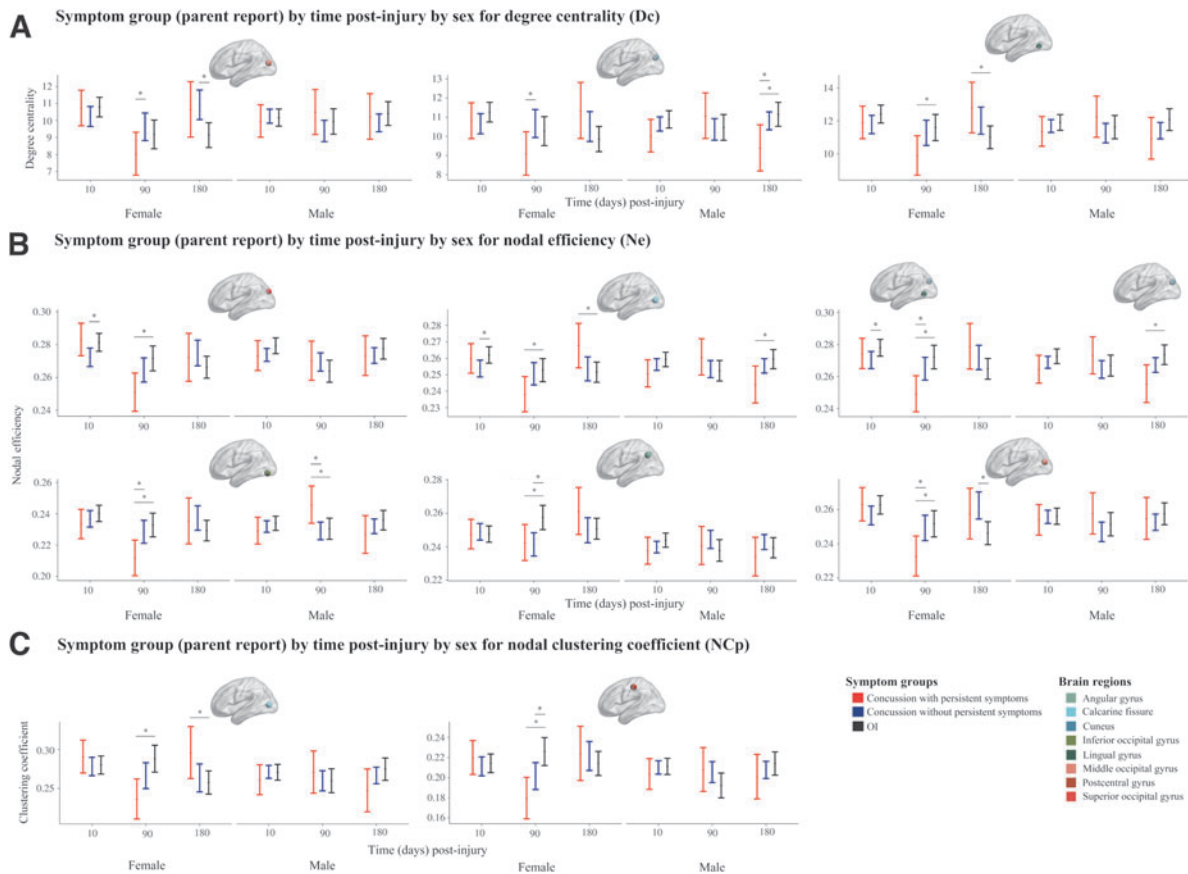


FIG. 5. Differences in regional (nodal) graph metrics among symptom groups based on parent report were moderated by time post-injury and biological sex. Graphs illustrate the moderating effects of time post-injury by sex for **(A)** degree centrality, **(B)** nodal efficiency, and **(C)** nodal clustering coefficient. For each graph, results are shown on the left for females and on the right males. Cohen's *d* and 95% confidence intervals are shown for robust effects

the first several weeks post-injury are small and are poor predictors of clinical outcomes in pediatric concussion.⁸³ However, data-driven analysis techniques (e.g., machine learning) have differentiated concussion from OI based on post-acute functional brain network metrics with high sensitivity and specificity,⁸⁴ and should be a focus of future research.

In contrast to post-acute differences, regional and local network alterations were more robustly demonstrated chronically (i.e., 3 or 6 months post-injury) after pediatric concussion, but varied by biological sex. Generally, chronic alterations (i.e., reduced clustering coefficient and nodal efficiency) suggested reduced segregation and connectivity, and were demonstrated by 3 months after concussion relative to OI. However, alterations emerged at 3 months post-injury in females but emerged at 6 months post-injury in males. Therefore, the effects of concussion on brain network functioning appear to take longer to emerge in males than in females, but additional

research is needed to determine whether the eventual resolution of differences currently observed in females occurs in males beyond 6 months post-injury.

Local network function changes also emerged 3 months after mild OI. A growing body of evidence has shown brain changes following OI.^{85,86} Changes in the OI group could reflect several factors, including the effects of pain and neuroinflammation.⁸⁷ A previous study showed white matter microstructure changes between 96 h and 3 months after OI in adolescents and young adults, whereby differences attenuated (reduced) over time relative to uninjured typical peers.²² Therefore, the inclusion of a third comparison group of uninjured, typically developing children in future studies could help to determine whether the currently demonstrated changes in the OI group reflect an attenuation or exacerbation of injury-related effects.

When considering that both older adolescents and girls have increased risk of protracted recovery after

concussion,^{3,88} these results particularly suggest that biological differences related to sex differences influence local and regional brain function changes following pediatric concussion. Females with concussion demonstrated increased local segregation (i.e., increased clustering coefficient and degree centrality) with hyperconnectivity (see Fig. 5) later in time, at 6 months post-injury. This is similar to the hyperconnectivity demonstrated in moderate to severe TBI¹⁶ and in adult females 6 months after mild TBI.⁸⁹ The pattern could indicate neural compensation in response to inefficient recruitment of regions that showed changes at 3 months post-injury. However, neural compensation may not be beneficial but could instead reflect microstructural or metabolic alterations,⁹⁰ especially considering that this pattern of results was particularly apparent in females with slow recovery following concussion (i.e., concussion with persistent symptoms). Future investigations of symptom trajectories, specifically examining clinical outcomes at 6 months post-injury in relation to brain imaging, would help to clarify whether the demonstrated pattern of hyperconnectivity following pediatric concussion in females is beneficial or detrimental.

Local network metrics were altered in regions of the visual network and several additional posterior brain regions after concussion relative to OI. Previous functional neuroimaging studies found altered posterior connectivity in adolescents after concussion within 1 week post-injury,⁹¹ including hyperconnectivity of the precuneus and right middle occipital gyrus.¹⁰ Reduced functional connectivity also was demonstrated in the right lateral parietal and supramarginal gyrus in adolescents with concussion relative to OI 1 month post-injury.¹¹ Similar functional network alterations have also been demonstrated in a recent magnetoencephalography study.⁸⁴ The etiology of functional changes after concussion is poorly understood. These results suggest that functioning of the visual network is particularly susceptible to the effects of concussion, with alterations especially apparent at later time periods post-injury. Mounting evidence from pre-clinical animal models of TBI indicates that axonal, but not neuronal, degeneration occurs within regions of the visual network, including several of those involved in this study, at later recovery times following concussion.⁹²

Limitations

The current study has some limitations. Participants were recruited from emergency departments and may not be representative of individuals who seek care in other settings^{22,93,94} or who do not seek medical attention.⁹⁵ The current sample consisted primarily of children from families of higher socioeconomic status and may not generalize to less advantaged populations. Although children with OI having any head trauma or signs or symptoms

of concussion were excluded from this group, the possibility of occult or sub-concussive brain injury cannot be entirely ruled out.^{96–98} However, we view the inclusion of an OI comparison group as a methodological strength of the current study. Specifically, including an OI group instead of a healthy control group can account for pre-morbid factors related to increased risk of childhood injury. It also can account for post-injury factors that could have influenced brain network function (e.g., pain, inflammatory response to injury, disrupted routine, and medical interactions),^{27–31} and for the psychosocial effects associated with a traumatic injury (e.g., acute emotional stress).²⁰ Therefore, we believe that the OI comparison group offers a clinically relevant and conservative comparison group for determining differences that are specifically related to concussion rather than to pre-morbid or general post-injury factors. Persistent symptom status was assessed based on a total symptom score at a single time point (i.e., 1-month post-injury). Future research should examine specific symptom subtypes and account for changes in symptom trajectories across time to better capture the highly variable and heterogeneous clinical phenotypes that can occur following pediatric concussion.^{3,4,9} Weighted adjacency matrices were based on the AAL-90 atlas, and were based on absolute correlation values following a proportional threshold. Different methodological choices (e.g., different atlas, motion correction) could influence results.⁹⁹

Conclusion

In this large prospective, longitudinal cohort study of the functional connectome in pediatric concussion as compared with mild OI, local network alterations in functional connectivity were apparent within the first few weeks post-injury and were followed by dynamic trajectories of changes in both global and local network metrics up to 6 months post-injury. Group differences were influenced by symptom persistence at 1 month post-injury, time post-injury, biological sex, and, to a lesser extent, age at injury. Although group differences were present acutely, the small to modest effect size of group differences and the moderating effects of sex likely would preclude effective clinical application for diagnosis in individual patients. These findings highlight the importance of contextualizing pediatric concussion and its outcomes within a neurodevelopmental framework that considers children's age at the time of injury and their biological sex. Future research should examine alterations in the visual network in relation to visual function and symptoms to determine whether the differences provide a marker of functional disruption.

Transparency, Rigor, and Reproducibility Summary

The study protocol was published shortly after the A-CAP study began.³³ The study protocol included a

general analytic plan, but the specific analysis plan for this article was not formally pre-registered. A sample size of 700 in the concussion group and 300 in the OI group was planned to provide adequate power to detect small effect sizes. A total of 6449 children were screened in the ED, 3051 were eligible, and 979 were consented and enrolled, of whom 12 withdrew or were later determined to be ineligible. The final data set included 918 scans from 585 children (post-acute $n=502$; 3 months post-injury $n=197$; 6 months post-injury $n=219$). Key inclusion criteria were based on an established definition. Data were collected by research staff blind to relevant characteristics of the participants. Participants were not informed of their metrics on the measures used in the current analyses. All surveys and questionnaires used in the study are available from the authors. The primary measures used in the current analyses have demonstrated acceptable psychometric characteristics. Missing data were reported in the text. De-identified data from this study are not available in a public archive but can be made available upon reasonable request from any qualified investigator, subject to a signed data access agreement and as allowable according to institutional ethics board standard, by emailing the senior author (K.O.Y.). Analytic code used to conduct the analyses presented in this study are not available in a public repository but may be available by emailing the corresponding author (A.L.W., A.I.O.). The authors agree to provide the full content of the manuscript on request by contacting the senior author (K.O.Y.).

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Authors' Contributions

All authors made substantial contributions to the conception or design of the work, or to the acquisition, analysis, or interpretation of data for the work; assisted in drafting the work or revising it critically for important intellectual content; provided final approval of the version to be published; and agree to be accountable for all aspects of the work and ensure that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. Specific contributions are as follows: A.I.O. was responsible for conception and design of the study and acquisition and analysis of data, and drafted a significant portion of the manuscript and figures; S.D. was responsible for conception and design of the study and acquisition and analysis of data; K.O.Y. was responsible for conception and design of the study and acquisition, analysis of data, and critical revision of the manuscript and drafted a significant portion of the manuscript and figures; S.B. was responsible for concep-

tion and design of the study, acquisition and analysis of data, and critical revision of the manuscript; K.G. was responsible for conception and design of the study, acquisition and analysis of data, and critical revision of the manuscript; N.A., M.H.B., C.B., B.B., W.C., M.D., S.D., Q.D., S.B.F., B.G.G., and J.G. were responsible for conception and design of the study and critical revision of the manuscript; C.L. was responsible for conception and design of the study, acquisition and analysis of data, and critical revision of the manuscript; A.A.L. was responsible for conception and design of the study and critical revision of the manuscript; R.Z. was responsible for conception and design of the study and critical revision of the manuscript; and A.L.W. was responsible for conception and design of the study and acquisition and analysis of data, and drafted a significant portion of the manuscript and figures.

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Supplementary Material

Supplementary Figure S1
Supplementary Table S1
Supplementary Table S2
Supplementary Table S3

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